

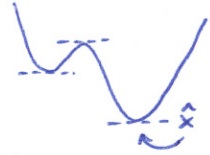
①

Matrix calculus

Minimizing a smooth function (unconstrained)

$$\hat{x} = \arg \min_x f(x) \implies f'(\hat{x}) = 0$$

↑ only necessary,
not sufficient!

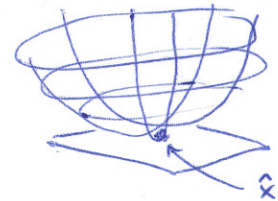


In multiple variables (again smooth + unconstrained)

$$f: \mathbb{R}^n \rightarrow \mathbb{R}. \quad \text{example, } n=2.$$

$$\hat{x} = \arg \min_x f(x) \implies \nabla f(\hat{x}) = 0 \quad \left(\text{or } \frac{d}{dx} f(x) = 0 \right)$$

↑ again only
necessary



$$\nabla f: \mathbb{R}^n \rightarrow \mathbb{R}^n \quad \nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}$$

Examples:

$$f(x) = c_1 x_1 + c_2 x_2 + \dots + c_n x_n \quad (\text{linear function})$$

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

written another way...

$$\boxed{\nabla_x (c^T x) = c}$$

Example #2 : (2-norm)

(2)

$$\frac{d}{dx}(\|x\|^2) = \frac{d}{dx} \left(\sum_{i=1}^n x_i^2 \right).$$

$$\frac{\partial}{\partial x_i} \left(\sum_{j=1}^n x_j^2 \right) = 2x_i \Rightarrow \frac{d}{dx}(\|x\|^2) = 2x.$$

Example #3 : (quadratic form)

$$f(x) = x^T Q x \quad (\text{usually assume } Q \text{ is symmetric}).$$

i.e. replace Q by $\frac{1}{2}(Q + Q^T)$.

$$\frac{d}{dx} x^T Q x = \frac{d}{dx} \sum_{i=1}^n \sum_{j=1}^n Q_{ij} x_i x_j$$

$$\frac{\partial}{\partial x_k} \sum_{j=1}^n \sum_{i=1}^n Q_{ij} x_i x_j = \begin{cases} i=j=k \rightarrow 2Q_{kk} x_k \\ i=k, j \neq k \rightarrow Q_{kj} x_j \\ j=k, i \neq k \rightarrow Q_{ik} x_i \end{cases}$$
$$= (Q + Q^T)x.$$

alternative proof:

$$\frac{d}{dx} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}^T \begin{pmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \\ q_{31} & q_{32} & q_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{d}{dx} \left\{ \begin{aligned} & q_{11}x_1^2 + q_{12}x_1x_2 + q_{13}x_1x_3 \\ & + q_{21}x_1x_2 + q_{22}x_2^2 + q_{23}x_2x_3 \\ & + q_{31}x_1x_3 + q_{32}x_2x_3 + q_{33}x_3^2 \end{aligned} \right\}$$

$$\frac{d}{dx_1} = 2q_{11}x_1 + q_{12}x_2 + q_{13}x_3 + q_{21}x_2 + q_{31}x_3$$

$$= [q_{11} \ q_{12} \ q_{13}]x + [q_{21} \ q_{21} \ q_{31}]x.$$

$$\frac{d}{dx_2} = [q_{21} \ q_{22} \ q_{23}]x + [q_{12} \ q_{22} \ q_{32}]x$$

$$\frac{d}{dx_3} = [q_{31} \ q_{32} \ q_{33}]x + [q_{13} \ q_{23} \ q_{33}]x$$

$$\left. \begin{aligned} & [q_{11} \ q_{12} \ q_{13}]x + [q_{21} \ q_{21} \ q_{31}]x \\ & [q_{21} \ q_{22} \ q_{23}]x + [q_{12} \ q_{22} \ q_{32}]x \\ & [q_{31} \ q_{32} \ q_{33}]x + [q_{13} \ q_{23} \ q_{33}]x \end{aligned} \right\} (Q + Q^T)x$$

$$= 2Qx \quad \text{if } Q = Q^T.$$

(3)

Deriving the normal equations

$$\underset{x}{\text{minimize}} \|y - Ax\|^2 \quad \text{where } y, A \text{ are given.}$$

$$\begin{aligned} \frac{d}{dx} \|y - Ax\|^2 &= \frac{d}{dx} (y - Ax)^T (y - Ax) \\ &= \frac{d}{dx} \left(x^T (A^T A) x - \underbrace{y^T A x - x^T A^T y}_{\text{the same!}} + y^T y \right) \\ &= \frac{d}{dx} \left(x^T (A^T A) x - 2 y^T A x \right) \\ &= 2(A^T A)x - 2A^T y \end{aligned}$$

set equal to zero: $\boxed{A^T A x = A^T y}$ (Normal equations!)

Quadratic forms and positive-definite matrices

4

if $Q \in \mathbb{R}^{n \times n}$ then a function of the form

$$f(x) = x^T Q x$$

is called a quadratic form. Can always assume Q is symmetric.

(quadratic form for Q is same as quadratic form for $\frac{1}{2}(Q+Q^T)$)

A matrix $Q \in \mathbb{R}^{n \times n}$ is positive definite if $x^T Q x > 0$ for all $x \neq 0$.

A matrix $Q \in \mathbb{R}^{n \times n}$ is positive semidefinite if $x^T Q x \geq 0$ for all x .

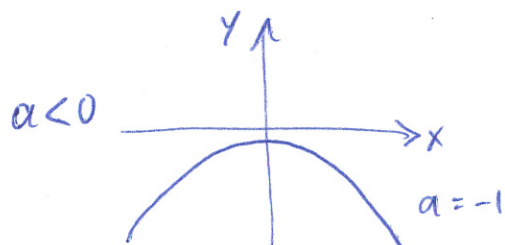
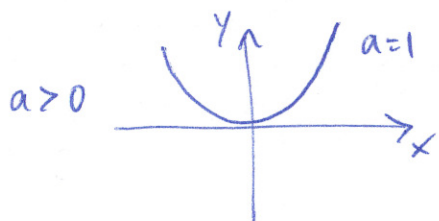
notation: $Q > 0$ or $Q \geq 0$

or $Q > 0$ or $Q \geq 0$

(not same as element-wise positive matrices!)

graphical

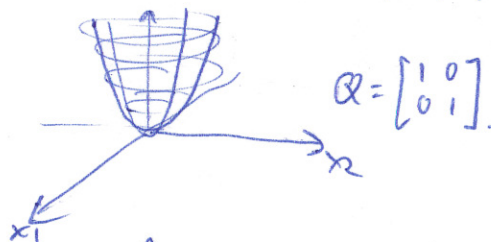
quadratic: $y = ax^2$



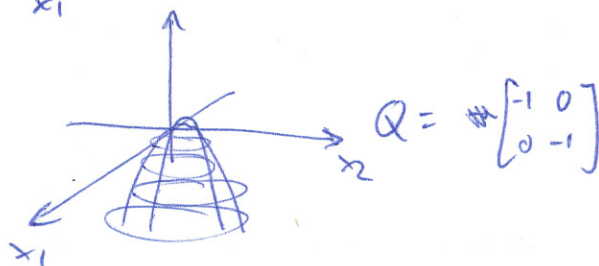
(only possibilities!)

quadratic form: $y = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^T \begin{pmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

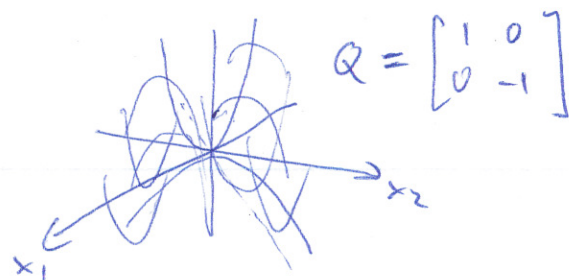
$Q > 0$



$Q < 0$



Q indefinite



(5)

properties of PSD matrices

- 1) if $P > 0$ and $Q > 0$ then $P + Q > 0$.
- 2) if $Q > 0$ then $aQ > 0$ for all $a > 0$.
- 3) if $P > Q$ and $Q > R$ then $P > R$
- 4) if $P > 0$ then $-P < 0$.
- 5) NOT TRUE if $P > 0, Q > 0$ then $PQ > 0$

equivalent definition

- (i) $Q > 0$
- (ii) $x^T Q x > 0 \quad \forall x \neq 0$.
- (iii) $\lambda_i > 0$ for all eigenvalue of Q .
- 6) if A is any matrix, $A^T A \geq 0$ and $AA^T \geq 0$.
if A has lin. indep. columns $\Leftrightarrow A^T A > 0$.
- 7) if $A > 0$ then A^{-1} exists.
- 8) if $A > B > 0$ then $B^{-1} > A^{-1}$.